

Fund Flow Risk and Cross-Sectional Asset Prices

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Abstract

This paper documents that stocks whose historical excess returns have co-moved more with mutual fund flows—have a higher fund flow beta (FFB)—earn higher future excess returns than otherwise similar stocks. We propose two hypotheses to explain this finding: First, funds may hedge the variance in their AUM by underweighing stocks with a high FFB, hence reducing the price of those stocks in equilibrium. Second, flow-induced trading in funds may give rise to a price pressure effect. Thus, stocks with a higher FFB have been differentially exposed to this price pressure. If the price pressure is asymmetric, the higher future excess return on these stocks may be a reversal effect. We find empirical support for our second hypothesis. Based on these findings, we construct a simple long-short strategy that earns an average monthly four-factor alpha of 0.5%.

Keywords: Asset pricing, mutual fund flows, price pressure, institutional trading

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1 Introduction

Classical asset pricing theories, like the Sharpe-Lintner CAPM, assume that all trades in the financial markets are carried out by investors who strive to maximize the expected mean-variance return of their investment portfolios. This is a strong assumption. While it may be valid—at least in a normative sense—with regard to retail investors, the same cannot be said about institutional investors; the latter type of investor does not have a utility function mapped over its own wealth and hence may face a very different maximization problem.

Consider the potential impact of institutional trading on asset prices. As of 2014, no less than 47% of the total outstanding equity in all US firms is held by US financial institutions. In comparison, only 37% of the equity is held by US households.¹ Under these circumstances, and considering the fact that there is ample evidence that the demand curves for stocks slope downward,² it is easy to see that we will never fully understand asset prices unless we also have a clear grasp of the incentives that drive institutional trading patterns.

In spirit of the above, this paper investigates the pricing of fund flow risk. Specifically, using a Fama-Macbeth regression procedure and controlling for the four Fama-French-Carhart factors, we find that stocks that have comoved more with historical, economy-wide fluctuations in fund flows earn higher abnormal returns. We propose two hypotheses to explain this finding:

Hypothesis 1 (the hedging hypothesis). Our first hypothesis is derived from the literature on agency-based asset pricing. A common belief in this field is that fund managers face a utility function that can be expressed as the return on their fund’s portfolio relative to the fund’s benchmark index.³ We suggest an alternative utility function that accounts for flow-based considerations. More concretely, under this hypothesis, fund managers hedge the variance in their assets under management by underweighing stocks that comove with fund flows, hence reducing the price of those stocks in equilibrium.

Hypothesis 2 (the price pressure hypothesis). Our second hypothesis is derived from the literature on flow-induced price pressure. This literature suggests that stocks held by open-end funds may be exposed to price pressure as funds scale up (down) their positions in response to sales (redemptions).

¹These figures are based on Federal Reserve’s Flow of Funds data. Most of the remaining balance is held by non-US entities.

²For example, Shleifer (1986) finds that stocks experience a persistent price increase upon being included in the S&P500 index. He attributes this effect to an outward shift in the stocks’ demand curves from purchases by index funds.

³See, for example, Brennan (1993), Cornell and Roll (2005), and Basak and Pavlova (2013).

Thus, under this hypothesis, we propose that stocks that have comoved more with historical fund flows have been differentially exposed to this price pressure. Further, given that the price pressure is asymmetric, the higher future abnormal return on these stocks may be a reversal effect.

Taking both hypotheses to the data, we reject the first but find strong support for the second hypothesis; First, controlling for common stock characteristics preferred by mutual funds, we find that mutual funds overweigh stocks that comove more with fund flows. This finding directly contradicts the hedging hypothesis. Second, consistent with the price pressure hypothesis, we show that funds hold relatively less (more) of stocks that have comoved more with past flows when those flows were negative (positive). Third, lending additional support for the price pressure hypothesis, we find that the premium on the fund flow factor is negatively correlated with lagged flows. This finding suggests that the price impact from historical comovements follows a reversal pattern. Moreover, we show that this effect is stronger when past flows have been negative, indicating that funds exert a larger price pressure when they are forced to sell off stocks due to redemptions than when they scale up their positions in response to positive flows. As a result, positive reversals dominate negative reversals, thus explaining why the risk premium on the fund flow factor is positively priced on average.

The contribution of this paper is threefold. First, it shows that stocks that have experienced larger comovements with economy-wide fluctuations in fund flows earn higher future excess returns, *ceteris paribus*. In doing so, it identifies a new risk factor that appears to have a significant explanatory power on asset prices.

Second, it shows that the pricing of fund flow risk cannot be explained by a desire from fund managers to underweigh stocks that are differentially exposed to fund flow risk. While this result runs counter to what one would expect from an agency-based perspective, it does lend some support for the supposition that the utility function that is most commonly assigned to fund managers within this area of research—the return on the fund portfolio in excess of the fund’s benchmark index—is well-specified.

Last, it shows that the pricing of fund flow risk is mainly the result of a reversal effect that arises due to flow-induced trading from mutual funds. This result corroborates a number of recent papers that find evidence of a price impact from liquidity-based trading in open-end funds, most notably Coval and Stafford (2007) and Lou (2012). Further, it augments the results from these papers by showing that the price impact is highly asymmetric along the axis of positive and

negative flows; specifically, reversals are bigger and more lagged following extreme negative flows.

This paper proceeds as follows: Section 2 describes our data and our methodology to estimate the stock betas and factor risk premia. Section 3 investigates the premium on the fund flow factor. Section 4 develops and tests our two main hypotheses; section 4.1 explores the hedging hypothesis, while section 4.2 explores the price pressure hypothesis. Section 5 concludes.

2 Data and Methodology

2.1 Data

Our fund flow data comes from the Investment Company Institute (ICI), which covers the majority of mutual funds on the US market. We use the data for funds classified as either hybrid funds or equity funds, since we are primarily interested in the behavior of fund managers that trade in equity. The data covers the period January 1984 to December 2014 and consequently includes a total of 374 monthly observations.

We use stock data from the Center for Research in Security Prices (CRSP), which covers the entire US equity market. The data includes closing prices, shares outstanding, and monthly returns. The return variable is adjusted for stock splits and dividends. We use only stocks that are listed on either NYSE, AMEX, or NASDAQ. In total, the number of firms in our sample period amounts to 19,338. The four Fama-French-Carhart factors are downloaded from the Kenneth French's website at Dartmouth. The data covers the period January 1984 to December 2014.

Fund holdings data is collected from the Thomson-Reuters s12 master file, which covers all US mutual funds and the majority of their US equity holdings.⁴ We use quarterly data and the sample period stretches from January 2000 through September 2014.

2.2 Estimation of beta coefficients

To standardize the fund flow data, for each month t , we compute the fund flow percentage (FFP) as aggregate flow, divided by the total market capitalization of

⁴The potential exclusions include, for example, small holdings (typically under 10,000 shares or \$200,000) and cases where there may be confidentiality issues.

all stocks in our sample, as in Warther (1995). Mathematically, for each month t :

$$FFP_t = \frac{flow_t}{\sum_i marketcap_{i,t}} \quad (1)$$

... where $flow_t$ is the aggregate fund flow to all hybrid and equity mutual funds in month t and $marketcap_{i,t}$ is the market capitalization of stock i in month t .

We use a rolling regression with time-varying betas to estimate the coefficients on the market, size, and value factors from Fama and French (1993), the momentum factor from Carhart (1997), and the fund flow percentage, defined as in (1). We estimate the coefficients for each stock i in every month t using the stock's past four years of monthly excess returns, i.e., from month $t - 47$ to t .⁵ Thus, at the end of each month t , we run the following regression for stock i and define the stock's fund flow beta (FFB) as the coefficient on FFP :

$$R_{i,s} - R_{f,s} = \alpha_i + \beta_{i,t}^M \times MKT_s + \beta_{i,t}^S \times SMB_s + \beta_{i,t}^H \times HML_s + \beta_{i,t}^U \times UMD_s + \beta_{i,t}^F \times FFP_s + \varepsilon_{i,s}, \quad s = t - 47, \dots, t \quad (2)$$

... where $R_{i,s}$ is the return of stock i in month s , $R_{f,s}$ is the risk-free rate (1 month treasury bill) in month s , MKT is the stock market excess return over the risk free rate, and SMB , HML , and UMD are spreads between portfolios sorted by market capitalization, book-to-market ratio, and prior 12-month average return, respectively.

To increase comparability, the resulting betas from (2) are scaled by their respective standard deviations. Furthermore, since we believe that market participants base their predictions of future betas on historical betas, we move each observation one month ahead for each coefficient for each stock i . This means that the beta estimated at the end of month t will be the grounds for investment decisions in the beginning of month $t + 1$.

2.3 Estimation of risk premia

We use a Fama-Macbeth regression (Fama and MacBeth 1973) to estimate the risk premia of the coefficients from (2): β^M , β^S , β^H , β^U , and β^F . Specifically, for each month t , we run cross-sectional regressions of the stocks' excess returns on their beta coefficients. In so doing, the resulting regression coefficients represent the pricing of risk for the corresponding risk factor. Mathematically, the regression

⁵To reduce noise in estimating FFB , we require that stocks have at least 36 non-missing traded returns during the 48-month estimation period.

we run is the following:

$$R_{i,t} - R_{f,t} = \lambda_i + \gamma_t^M \times \beta_{i,t}^M + \gamma_t^S \times \beta_{i,t}^S + \gamma_t^H \times \beta_{i,t}^H + \gamma_t^U \times \beta_{i,t}^U + \gamma_t^F \times \beta_{i,t}^F + \varepsilon_{i,t} \quad (3)$$

... where $R_{i,t}$ is the return of stock i in month t , $R_{f,t}$ is the risk-free rate (1 month treasury bill) in month t , $\beta_{i,t}^M$ is the market coefficient for stock i in month t , $\beta_{i,t}^S$ is the size coefficient for stock i in month t , $\beta_{i,t}^H$ is the value coefficient for stock i in month t , $\beta_{i,t}^U$ is the momentum coefficient for stock i in month t , and $\beta_{i,t}^F$ is the FFB for stock i in month t .

3 Results

The first row in Table 1 reports results from (3) for the whole sample period. Most notably, the premium on the fund flow factor, γ^F , is positive and significant at the 5% level. The coefficient suggests that a one standard deviation increase in the fund flow beta of a stock increases its excess return by 0.23 percentage points per month, all else equal.

Rows two through six report results for the same regression for five smaller time periods. Notice that the period 2006–2010 is the only period that shows a significant coefficient on FFB. Similarly, while insignificant, the period 2001–2005 shows a positive coefficient that is notably larger than in the other periods, except 2006–2010. These two periods have in common that the stock market experienced large crashes: the bursting of the dot-com bubble in 2000 and the financial crisis of 2007–2008. Ergo, these results indicate that the risk premium on the fund flow factor is higher after extreme downturns in the market.

Interestingly, we find no support for a positive risk premium on either the market factor, the size factor, or the value factor, which goes against previous research. Similarly, the pricing of the momentum factor is significantly negative, on the 5% level, which is not consistent with previous findings. However, this finding can perhaps be explained by our estimation methodology, where we look at the historical betas under a four-year period for each stock. Thus, stocks that load highly on the momentum factor have likely experienced a positive price development for a rather long time. Therefore, it is possible that the negative premium on the momentum factor in fact captures long-term price reversals, similar to the findings in De Bondt and Thaler (1985).

Figure 1 plots the cumulative premium on the fund flow factor during the sample period. Again, we clearly see that the risk premium is positive and that

Table 1: Cumulative risk premia

For six different periods, this table reports results for (3):

$$R_{i,t} - R_{f,t} = \lambda_i + \gamma_t^M \times \beta_{i,t}^M + \gamma_t^S \times \beta_{i,t}^S + \gamma_t^H \times \beta_{i,t}^H + \gamma_t^U \times \beta_{i,t}^U + \gamma_t^F \times \beta_{i,t}^F + \varepsilon_{i,t}$$

... where $R_{i,t}$ is the return of stock i in month t , $R_{f,t}$ is the risk-free rate (1 month treasury bill) in month t , $\beta_{i,t}^M$ is the market coefficient for stock i in month t , $\beta_{i,t}^S$ is the size coefficient for stock i in month t , $\beta_{i,t}^H$ is the value coefficient for stock i in month t , $\beta_{i,t}^U$ is the momentum coefficient for stock i in month t , and $\beta_{i,t}^F$ is the FFB for stock i in month t . Betas have been normalized by their standard deviations. Returns are denoted in percent. The reported coefficients are the means of the coefficients during the corresponding periods. t-stats are reported in parentheses.

	$\bar{\gamma}^F$	$\bar{\gamma}^M$	$\bar{\gamma}^S$	$\bar{\gamma}^H$	$\bar{\gamma}^U$	Constant
1987-2015	0.231** (2.12)	0.111 (0.84)	-0.001 (-0.01)	0.130 (0.85)	-0.177** (-2.28)	0.738*** (3.95)
1987-1995	-0.055 (-0.51)	0.039 (0.23)	0.057 (0.44)	0.099 (0.75)	-0.050 (-0.47)	0.570* (1.72)
1996-2000	0.086 (0.48)	-0.038 (-0.16)	-0.019 (-0.04)	0.030 (0.06)	-0.176 (-1.06)	0.770* (1.74)
2001-2005	0.348 (1.32)	-0.050 (-0.12)	0.001 (0.01)	0.114 (0.18)	-0.293 (-1.31)	1.552*** (4.05)
2006-2010	0.976** (2.23)	0.864** (2.26)	-0.030 (-0.19)	0.394 (1.56)	-0.411 (-1.56)	-0.491 (-0.91)
2011-2015	-0.041 (-0.20)	-0.288 (-1.00)	-0.069 (-0.45)	0.016 (0.13)	-0.019 (-0.23)	1.581*** (4.83)

there is a significant jump after the financial crisis of 2007–2008. In fact, this jump represents more than 50% of the cumulative premium on the fund flow factor during the sample period.⁶

Overall, these results strongly support that there is a positive risk premium on the fund flow factor. As a—to the best of our knowledge—previously undocumented fact in the literature, this finding warrants further investigation. Thus, in the following section, we propose two hypotheses that try to explain the positive risk premium on the fund flow factor.

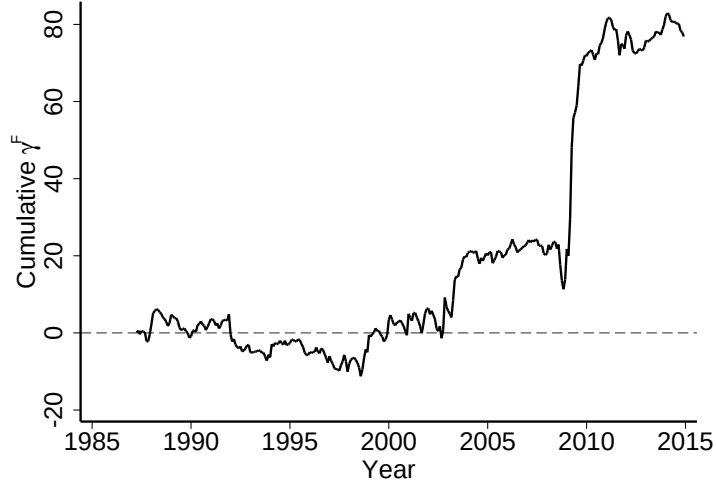


Figure 1: Cumulative risk premium on the fund flow factor

This figure plots the cumulative risk premium on the fund flow factor for the full sample period. The cumulative premium is the sum of the monthly γ_t^F in (3):

$$R_{i,t} - R_{f,t} = \lambda_i + \gamma_t^M \times \beta_{i,t}^M + \gamma_t^S \times \beta_{i,t}^S + \gamma_t^H \times \beta_{i,t}^H + \gamma_t^U \times \beta_{i,t}^U + \gamma_t^F \times \beta_{i,t}^F + \varepsilon_{i,t}$$

... where $R_{i,t}$ is the return of stock i in month t , $R_{f,t}$ is the risk-free rate (1 month treasury bill) in month t , $\beta_{i,t}^M$ is the market coefficient for stock i in month t , $\beta_{i,t}^S$ is the size coefficient for stock i in month t , $\beta_{i,t}^H$ is the value coefficient for stock i in month t , $\beta_{i,t}^U$ is the momentum coefficient for stock i in month t , and $\beta_{i,t}^F$ is the FFB for stock i in month t . Betas have been normalized by their standard deviations. Returns are denoted in percent. Thus, γ_t^F is the estimated increase in a stock's excess return, denoted in percent, for every one standard deviation increase in its FFB, holding the loadings on the other factors constant.

4 Hypotheses

4.1 The hedging hypothesis

The effect of institutional trading on classical asset pricing models has been discussed in a number of previous works. One of the earliest models that describes

⁶Section 4.2 specifically investigates this response to large downturns in the market.

the effect is presented in Brennan (1993). He proposes a model where institutional investors have absolute risk aversion with respect to their performance relative to some benchmark. Under this assumption, he shows that institutional investors will tilt their holdings toward stocks that comove more with the benchmark. Therefore, in equilibrium, those stocks will have higher prices and lower expected returns.

To test his model, Brennan uses the S&P500 index as a proxy for the benchmark index and finds only limited support. However, in a follow-up article, Brennan and Li (2008) find strong support for this model and conclude that “stocks whose returns covary more with the idiosyncratic part of the S&P500 return have significantly lower returns”.

Basak and Pavlova (2013) build upon Brennan’s results by introducing a similar model, but where agents have relative risk aversion. Their model has similar implications and explains phenomena such as index inclusion effects and asset class effects.

Finally, Cornell and Roll (2005) propose a model for an economy with only institutional investors. These investors are either active or passive and, like Brennan, they assume that the active managers are concerned with their return relative to a benchmark index. Using this model, they show that the CAPM is not a sufficient model in a world where investors delegate their investment decisions to institutional investors. Neither Basak and Pavlova nor Cornell and Roll test their models empirically.

In this section, we test if the positive premium on the fund flow factor is related to the utility function of fund managers and the hedging motives this utility function entails.

Fund managers’ utility function

The implied utility function in Brennan (1993) is:

$$U(R_{p,i} - R_{b,i}) = \begin{cases} (1 - e^{-a_i(R_{p,i} - R_{b,i})})/a_i & a_i \neq 0 \\ R_{p,i} - R_{b,i} & a_i = 0 \end{cases} \quad (4)$$

... where $R_{p,i}$ is the return on fund i ’s portfolio, $R_{b,i}$ is the return on the fund’s benchmark portfolio, and a_i is the agent’s risk aversion. We customize this utility function to include fund managers’ consideration of fund flow.

Assume that managers are concerned with maximizing their assets under management (AUM). This is a reasonable proposition since compensation to managers is largely dependent on the value of their portfolio. Furthermore, more AUM is likely positive for the reputation of a fund manager. Now, consider that AUM

increases due to two factors: a positive fund return and a positive fund flow. This implies that managers are concerned with their portfolio return and the fund flow to their fund.

With the above considerations in mind, we propose the following utility function for fund managers:

$$U(R_{p,i} + g_{ff,i}) = \begin{cases} (1 - e^{-a_i(R_{p,i} + g_{ff,i})})/a_i & a_i \neq 0 \\ R_{p,i} + g_{ff,i} & a_i = 0 \end{cases} \quad (5)$$

... where $R_{p,i}$ is the return on fund i 's portfolio, $g_{ff,i}$ is the growth in AUM related to fund flow for fund i , and a_i is the agent's risk aversion.

The asset pricing implications of (5) are developed in the same manner as for (4) in Brennan (1993). We consider a single period economy with two types of investors: individual investors who are mean-variance optimizers, and fund managers with a utility function as in (5). Asset returns are multivariate normal and agents are constrained to hold all of their wealth in equities. Under these assumptions, and given that we use FFP from (1) as the aggregate proxy for $g_{ff,i}$ (just as Brennan uses S&P500 as the aggregate benchmark proxy), Brennan shows that solving the maximization problem for fund managers provides the following asset pricing model:

$$E(R_i) = a + b \times \beta_i^M + c \times \beta_i^F \quad b > 0 \quad \text{and} \quad c > 0 \quad (6)$$

... where $E(R_i)$ is the expected return of stock i , β_i^M is the market beta of stock i , and β_i^F is the FFB of stock i .

Since the coefficient on FFB is positive in (6),⁷ this model explains the observed positive risk premium on the fund flow factor. This result is intuitive. A stock's comovement with FFP is captured by its FFB. Since managers want to maximize AUM but are risk averse, they naturally want to maximize their return, while minimizing their exposure to FFB. Per definition, having a high exposure to FFB leads to large downturns in AUM when fund flows are negative and vice versa when they are positive. Consequently, a high exposure to FFB increases the variance of AUM and is thus not desirable for fund managers from a mean-variance optimization perspective. This leads to the following proposition:

Fund holdings *Fund managers underweigh (overweigh) investments in high (low) FFB stocks. Thus, stocks with a high (low) FFB will be underpriced (over-*

⁷Notice that in Brennan (1993) this coefficient is negative. The reason for this is easily observed if we compare (4) with (5): The sign is positive for $g_{ff,i}$ while it is negative for $R_{p,i}$.

priced) in equilibrium, explaining the positive risk premium on the fund flow factor.

In the following section, we empirically test this proposition.

4.1.1 Fund holdings

To investigate the hedging hypothesis, we look directly at the holdings of mutual funds and how they are affected by FFB, with a slightly modified approach from Falkenstein (1996). Using quarterly fund holdings data, we construct a variable that represents the share ownership of stock outstanding held by the entire sample of mutual funds. Mathematically, for each quarter t and each stock i :

$$own_{i,t} = \sum_{m=1}^M \frac{\text{shares owned by fund } m \text{ at time } t}{\text{shares outstanding at time } t} \quad (7)$$

... where M is the total number of mutual funds in our sample.

If all asset-demanders held similar portfolios, then the mutual fund industry as a whole would own equal proportions of each stock. That being the case, if $own_{i,t}$ takes on a lower value for a particular stock than the industry average, the mutual fund industry in aggregate is averse to holding that stock. Using this variable, we can therefore test if mutual funds are averse to holding high FFB stocks.

To focus solely on the effect of FFB on mutual funds' holdings could lead to false conclusions, however. Falkenstein (ibid.) documents a number of stock characteristics that affect mutual funds' investment decisions. If FFB is correlated with any of these characteristics, our results will likely be confounded by the fact that other characteristics than FFB affect the holdings of mutual funds. Because of this, we control for a number of stock characteristics that have previously been showed to affect mutual funds' stock preferences. These characteristics are listed in Table 2 along with the result from the regression of the form:

$$y_{i,t} = \beta_1 \times x_{1,i,t} + \beta_2 \times x_{2,i,t} + \dots + \beta_n \times x_{n,i,t} + a_t + \varepsilon_{i,t} \quad y_{i,t} = \ln(1 + own_{i,t}) \quad (8)$$

... where $own_{i,t}$ is defined as in (7), $x_{1,i,t}, \dots, x_{n,i,t}$ are stock characteristics, specifically: FFB, standard deviation, variance, previous quarter return, liquidity, price, market capitalization, market coefficient, size coefficient, value coefficient, and momentum coefficient, and a_t captures time fixed effects. Control variables and the dependent variable are logarithmized to avoid outliers dominating the results. We use robust standard errors to account for heteroscedasticity and cluster stock identifiers to account for intra-cluster correlation.

Table 2 reports results for two versions of (8), with different control variables. Both regressions show a significantly positive coefficient on FFB, on the 1% level. Specifically, for each one standard deviation increase in FFB, mutual funds seem to increase their holdings of that particular stock by 0.4%–0.6%. Granted, that’s not a large increase. It is, however, significant enough for us to conclude that fund managers *do not* underweigh stocks with a high FFB, contrary to what we would expect under our first hypothesis. These results therefore completely contradict our first hypothesis, and we find no support for fund managers’ hedging behavior explaining the positive risk premium on the fund flow factor.

Table 2: Regression analysis: test of fund managers' hedging motives

This table presents results for the regression analysis testing the effect of fund flow beta on mutual fund holdings. The regression is of the form:

$$y_{i,t} = \beta_1 \times x_{1,i,t} + \beta_2 \times x_{2,i,t} + \dots + \beta_n \times x_{n,i,t} + a_t + \varepsilon_{i,t} \quad y_{i,t} = \ln(1 + own_{i,t})$$

... where $own_{i,t}$ is defined as

$$own_{i,t} = \sum_{m=1}^M \frac{\text{shares owned by fund } m \text{ at time } t}{\text{shares outstanding at time } t}$$

... where M is the total number of mutual funds in our sample. $x_{1,i,t}, \dots, x_{n,i,t}$ are stock characteristics, specifically: Stdev and variance are the standard deviation and variance of monthly returns for the prior 36 months. Ret(-1) is the return during the previous quarter. Liq is liquidity defined as the previous month's volume divided by shares outstanding at end of month. Size is the market capitalization. CAPM beta is the market beta. Column 2 presents the results including controls for stock correlation with the Fama-French-Carhart factors SMB, HML, and UMD. a_t captures time fixed effects. The sample period stretches from Jan 2000 through Sep 2014. Coefficients are denoted in percent. Robust standard errors were used to adjust for heteroscedasticity and observations were clustered based on stock identifiers to account for intra-cluster correlation. t-stats are reported in parentheses.

	Dependent variable: $\ln(1 + own_{i,t})$	
	(1)	(2)
Fund flow beta	0.392*** (5.56)	0.647*** (8.40)
Stdev	0.221 (0.09)	-6.996** (-2.50)
Variance	-5.645 (-1.64)	-0.701 (-0.18)
Ret(-1)	-2.074*** (-23.51)	-2.011*** (-22.93)
$\ln(liq+1)$	10.476*** (16.97)	10.217*** (16.58)
$\ln(prc+1)$	2.421*** (9.65)	2.498*** (9.73)
$\ln(size)$	12.792*** (18.09)	11.666*** (16.27)
$[\ln(size)]^2$	-0.273*** (-15.79)	-0.246*** (-14.04)
CAPM beta	0.565*** (7.29)	0.905*** (10.24)
SMB beta		0.944*** (12.75)
HML beta		-0.157** (-2.15)
UMD beta		-0.130* (-1.91)
Observations	228,289	228,289
R-squared	0.404	0.408

4.2 The price pressure hypothesis

Flow-driven price pressure has been studied to some extent in previous works. Interestingly, the earliest research on this topic only finds weak support for the existence of price pressure. Notably, Warther (1995) and Edelen and Warner (2001) both find evidence of a positive relationship between market returns and contemporaneous flows. However, they do not find any evidence of a negative relationship between market returns and lagged flows, as would be expected from price reversals if flows indeed are causative of (temporary) price shocks.

Nevertheless, recent papers have been more supportive of the existence of price pressure. For example, Teo and Woo (2004) group funds by their investment objective (e.g., one group consisting of all growth stock funds, etc.) and find that flows to such groups negatively predict future returns on the stocks held by the group. On the stock level, Coval and Stafford (2007) construct a measure that captures the degree to which a stock has been exposed to flow-constrained buys minus flow-constrained sells and find that it is negatively associated with subsequent returns. Similar results are achieved by Lou (2012) using a somewhat different measure of the amount of constrained trades in the stock.

Having rejected the hedging hypothesis in section 4.1, this section tests if the positive premium on the fund flow factor instead is related to a price pressure effect from flow-induced trading. The proposed mechanism and its testable implications (proposition 1–3) will be described in the next section. Section 4.2.1 to 4.2.3 show that the propositions hold in the data, and section 4.2.4 develops a trading strategy that takes advantage of these findings in order to demonstrate their economic significance.

Propositions

Our tests of the hedging hypothesis in section 4.1.1 showed that mutual funds have a *preference* for stocks that load highly on the fund flow factor. This finding suggests a positive relationship between fund flow betas and the extent to which a stock is traded by mutual funds. Therefore, it is conceivable that a higher loading on the fund flow factor may entail a higher exposure to price pressures from flow-induced trading in mutual funds. Indeed, if the price impact occurs immediately⁸ in response to flows (or if flows are autocorrelated), it may well be the case that the comovement does not capture a specific preference of stocks in mutual funds

⁸As we are looking at monthly data, ‘immediately’ is in this context identical to ‘within the same month.’

per se but is itself a direct result of flow-induced price pressures. This argument leads to our first proposition:

Proposition 1 (fund holdings). *High fund flow beta stocks are stocks that have been heavily sold (bought) by funds in response to negative (positive) flows. Thus, funds hold relatively less (more) of high fund flow beta stocks when past flows have been negative (positive).*

Contemplate the implication of the above proposition. If stocks that load highly on the fund flow factor have been heavily sold or bought by open-end funds and, for this reason, have been exposed to price pressure, such stocks will, on average, be underpriced (overpriced) following periods of negative (positive) flows. Hence, presuming that stock prices revert to their fundamentals in the long run, high fund flow beta stocks should outperform low fund flow beta stocks following periods of negative flows while the opposite should be the case following periods of positive flows. This dynamic is illustrated in Figure 2.A, and it leads to our second proposition:

Proposition 2 (relationship to lagged flows). *High fund flow beta stocks earn higher (lower) abnormal returns following periods of negative (positive) flows. In other words, the premium on the fund flow factor is negatively related to lagged flows.*

However, the mere existence of price pressure from flow-induced trading is not a sufficient criterion for the risk premium on the fund flow factor to be positively priced on average. Specifically, in order for the price pressure hypothesis to hold, high fund flow beta stocks must, on average, be underpriced. In other words, the price pressure must be asymmetric—that is, the absolute price impact from all negative past flows must be greater than the price impact from all positive past flows. This dynamic is illustrated in Figure 2.B, and it leads to our third—and last—proposition:

Proposition 3 (asymmetric price pressure). *High fund flow beta stocks are, on average, underpriced. Thus, the absolute value of the risk premium on the fund flow factor is greater when past flows have been negative compared to when past flows have been positive.*

There are at least two good reasons to expect proposition 3 to hold: First, funds are likely much less constrained in their trading activities when they face positive flows compared to when they face negative flows. In particular, when a

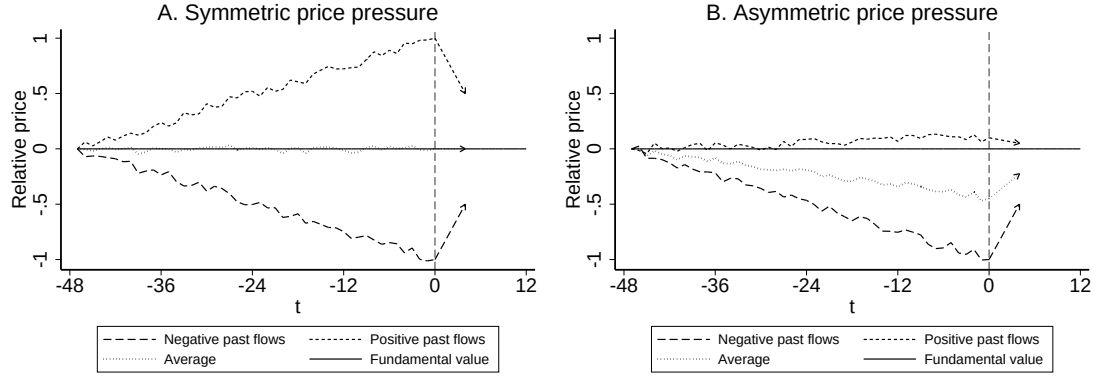


Figure 2: Schematic illustration of the price pressure hypothesis

This figure demonstrates the mechanisms of the price pressure hypothesis schematically. The dashed and dotted lines illustrate the price development of a representative stock that has been exposed to price pressure from either positive or negative flows during the estimation period of the betas (and thus load highly on the fund flow factor). The arrows indicate the expected reversal patterns (unconditioned on post-estimation-period flows). The x-axes show the month relative to the end of the estimation period of the betas. Precisely, $t = -47$ to $t = 0$ is the estimation period of the betas, and $t = 1$ is the estimation period of the risk premia. The y-axes show the price of the stock relative to its fundamental value. For illustrative clarity, the figure assumes that the stock is neither under- nor overpriced at the beginning of the estimation period. Furthermore, reversals are assumed to be linear and to take a constant amount of time.

In panel A, it is assumed that positive flows cause as much price pressure upward as negative flows cause price pressure downward. As a result, presuming that negative flows are of the same magnitude and frequency as positive flows, the stock will, on average, neither be over- nor underpriced at $t = 0$. Therefore, we would expect the average reversal at $t = 1$ —and hence the premium on the fund flow factor—to be zero.

In panel B, it is assumed that the price pressure from negative flows dominates the price pressure from positive flows. Thus, in contrast to panel A, the stock will, on average, be underpriced at $t = 0$. Consequently, we would expect the average reversal at $t = 1$ —and hence the premium on the fund flow factor—to be positive.

fund receives money, it need not invest it immediately or in any particular set of securities. Thus, if it perceives some securities to be overpriced, it has some leeway in choosing not to invest in those securities. In contrast, when a fund faces redemptions, it may be *forced* to sell a *particular set* of securities (its current holdings), even when those securities are underpriced.⁹ Adding to this effect, many funds only hold highly correlated, category-specific assets; for example, some funds only invest in stocks within a limited set of industry classes. Consequently, when such funds face severe outflows, the flow-induced price pressure is likely to be systemic to the entire fund portfolio, making it impossible to meet redemptions by only selling off assets that have not been exposed to any price pressure.

Second, as illustrated in Figure 3, the distribution of flows has a long negative tail. This observation is important because flow-induced price pressures should, realistically, chiefly be an outlier phenomenon. Specifically, because flow-based constraints sometimes impel funds to accept loss-making trades, they will seek to develop strategies to mitigate these constraints. However, such strategies will likely break down in the face of extreme flows, thus giving rise to price pressure. Consequently, since the distribution of flows is negatively skewed, negative price pressures should dominate positive dittos.

For example, one common strategy employed by funds in order to mitigate the impact of flows on their trading activities is to keep a cash reserve (see, for example, Coval and Stafford (2007)). Thus, when a fund faces redemptions (sales), it simply contracts (expands) the cash reserve to meet investor demands instead of adjusting any of its stock holdings. However, when the fund faces extreme, negative flows, the cash reserve will get depleted, and the fund may be forced to sell off underpriced assets, thus producing negative price pressure. Conversely, when the fund faces extreme, positive flows, the low-yielding cash reserve will get too big, and the fund may feel compelled to buy overpriced assets so as to match the equity share in peer funds or benchmark indices. Critically, then, because of the distribution of flows, the former scenario appears to be much more likely than the latter scenario.

In sum, considering the two reasons for asymmetry suggested above, it seems as if the price pressure hypothesis implies that the risk premium on the fund flow factor should respond favorably and be sensitive only to extreme negative flows. We will investigate this supposition in some further depth in section 4.2.3. For now, however, we note that the leftmost observations in Figure 3 closely preceded

⁹Note that, because of short selling constraints, it is not possible for mutual funds to raise capital by short selling stocks that they perceive to be less underpriced than any of the securities that they currently hold.

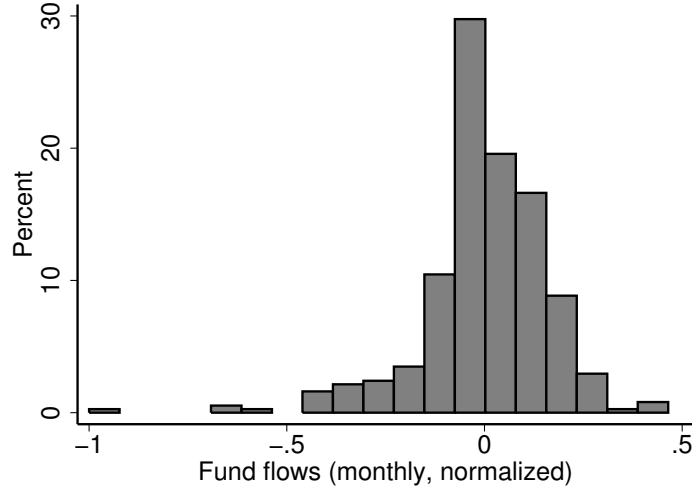


Figure 3: Distribution of monthly fund flows

This figure shows the distribution of fund flows to US hybrid and equity funds during the sample period, 1984–2015. The data source is the same as in section 2.1. The flows have been normalized by first subtracting the mean of flows and then dividing the resultant variable by its absolute max. The two leftmost clusters of observations occurred in connection to the 2007–2008 financial crisis and the dot-com bubble.

the two major jumps in the cumulative risk premium on the fund flow factor—the first jump occurred in the aftermath of the dot-com bubble and the second occurred following the 2007–2008 financial crisis (refer back to Figure 1).

Moving forward, the following sections examine the validity of the price pressure hypothesis by testing its main ramifications, proposition 1–3, empirically.

Unexpected fund flows

We base our tests of proposition 1–3 on *unexpected* fund flows.¹⁰ The reason for this is as follows: Since fund flows are highly autocorrelated, we expect that fund managers are, to some extent, able to predict future flows. Furthermore, it seems likely that fund managers can accommodate some amount of redemptions and sales by adjusting their cash reserves in anticipation of flows in order to mitigate price impacts. Because of this, we believe the effects of price pressure are more prominent when we look at unexpected flows—the part of fund flows that managers cannot predict based on previous flows.

To construct a variable that captures the unexpected part of fund flows, we adopt a similar approach to Warther (1995). We use an autoregressive model with three lags to predict future fund flows based on previous fund flows and define the

¹⁰The results are similar but somewhat less significant if we consider total (i.e., the sum of expected and unexpected) fund flows instead.

residual as the unexpected fund flow. The model is specified as follows:

$$FFP_t = c + \sum_{i=1}^3 \varphi_i FFP_{t-i} + \varepsilon_t \quad (9)$$

We define FFP_t^U (unexpected fund flow) as ε_t in (9).

4.2.1 Test of proposition 1 (fund holdings)

To test proposition one, we again look at mutual funds' holdings and how they are affected by FFB. However, unlike our earlier test, we take previous flows into consideration.

Based on FFP_t^U , we create a dummy variable, $D_t^{FFP^U}$, that takes on the value one when unexpected flows have been positive on average the past three months and zero otherwise. That is:

$$D_t^{FFP^U} = \begin{cases} 0 & \text{if } \sum_{s=t-2}^t FFP_s^U < 0 \\ 1 & \text{otherwise} \end{cases} \quad (10)$$

Finally, we create an interaction term that captures the effect of FFB on mutual funds' holdings, when the 3-month previous unexpected fund flows have been positive. This variable is simply the previously constructed dummy variable times FFB:

$$I_{i,t}^{FFP^U FFB} = D_t^{FFP^U} \times FFB_{i,t} \quad (11)$$

Using these variables, we can test our hypothesis by regressing the share of shares outstanding owned by mutual funds on fund flow beta, the dummy variable and the interaction term. The coefficient on the interaction term will then capture the isolated effect of mutual fund trading in high FFB stocks when previous flows have been positive. We run the following regression:¹¹

$$\ln(1 + own_{i,t}) = \beta_{1,i} \times FFB_{i,t} + \beta_{2,i} \times I_{i,t}^{FFP^U FFB} + a_t + \varepsilon_{i,t} \quad (12)$$

... where $own_{i,t}$ is defined as in (7), $FFB_{i,t}$ is defined as in (2), $I_{i,t}^{FFP^U FFB}$ is defined as in (11), and a_t captures time fixed effects. We use robust standard errors to account for heteroscedasticity and cluster stock identifiers to account for intra-cluster correlation.

¹¹Note that we do not include the dummy variable in the regression since its effect is captured in full by the time fixed effects.

Table 3 reports results from (12). Under this hypothesis, we would expect the coefficient on the interaction term to be positive, since that would indicate that mutual funds indeed hold more (less) high FFB stocks when previous unexpected flows have been positive (negative). As we can see, the coefficient is significantly positive on the 1% level. Specifically, if previous unexpected flows have been positive, mutual funds will hold 0.24 percentage points more of a stock for each one standard deviation increase in FFB for that stock, compared to if previous flows had been negative. While this is not a large increase, it does indicate that there might be some truth to our price pressure hypothesis.

Table 3: Regression analysis with interaction term: test of price pressure effect

This table presents results for the regression analysis testing the effect of fund flow beta, conditioned on previous unexpected fund flows:

$$\ln(1 + own_{i,t}) = \beta_{1,i} \times FFB_{i,t} + \beta_{2,i} \times I_{i,t}^{FFP^U FFB} + a_t + \varepsilon_{i,t}$$

... where $own_{i,t}$, fund holdings, is defined as the share ownership of stock outstanding held by the entire sample of mutual funds. $FFB_{i,t}$, fund flow beta, measures the comovement between a stock's excess returns and flows to mutual funds. $I_{i,t}^{FFP^U FFB}$, the past flow-FFB interaction term, captures the effect of FFB when unexpected fund flows have been positive the past three months. a_t captures time fixed effects. The sample period stretches from Jan 2000 through Sep 2014. Coefficients are denoted in percent. Robust standard errors were used to adjust for heteroscedasticity and observations were clustered based on stock identifiers to account for intra-cluster correlation. t-stats are reported in parentheses.

Dependent variable: $\ln(1 + own_{i,t})$	
Fund flow beta	-1.599*** (-18.70)
Past flow-FFB interaction term	0.242*** (6.40)
Observations	228,289
R-squared	0.094

The cautious reader may notice that the coefficient on FFB is negative in this regression, which would support our previously rejected hedging hypothesis. However, note that this is due to us not including the control variables for stock characteristics in this regression. The rationale behind that decision is that we here want to focus solely on the interaction between FFB, previous unexpected flows, and fund holdings. In results not shown, we find that these results hold when we include the stock characteristics from (8) and that the coefficient on FFB then is positive. Furthermore, these results hold when we control for firm fixed effects. Overall, these results support proposition one.

4.2.2 Test of proposition 2 (relationship to lagged flows)

Our second proposition states that the risk premium on the fund flow factor is negatively correlated with unexpected fund flows in preceding periods. This is an easily testable proposition. To test it, we regress the fund flow risk premium on contemporaneous and lagged unexpected fund flows:

$$\gamma_t^F = \beta_0 + \beta_1 \times FFP_t^U + \beta_2 \times \sum_{s=t-3}^{t-1} FFP_s^U + \dots + \beta_6 \times \sum_{s=t-18}^{t-16} FFP_s^U + \varepsilon_t \quad (13)$$

... where γ_t^F is the risk premium on the fund flow factor in month t and FFP_t^U is the unexpected fund flow, defined as in (9), in month t .

Table 4 column 1 estimates (13) on the entire sample of time periods, while columns 2 and 3 estimate (13) on two collectively exhaustive subsamples: one in which the cumulative unexpected fund flows over the prior 18 months were negative, and one in which they were positive.

Three things are of note here. First, we see a clear pattern in all three columns: The fund flow risk premium is negatively correlated with lagged unexpected fund flows. In the full sample and the negative past flow sample, there is a significant negative correlation between the fund flow risk premium and unexpected flows in the five preceding quarters, except the first preceding quarter. For the positive past flow sample, there is significant negative correlation between the fund flow risk premium and unexpected flows in the previous quarter, but not any other quarters. These results align with our second proposition.

Second, the fund flow risk premium is positively correlated with contemporaneous unexpected flows. This indicates that the fund flow risk premium follows the trading pattern in mutual funds—if mutual funds face redemptions (sales) due to negative (positive) unexpected flows, they are forced to sell off (buy) stocks and the contemporaneous fund flow risk premium is thus negative (positive)—lending more support to the price pressure hypothesis.

Third, the price pressure is larger and is exerted during a longer period when unexpected flows have been negative, compared to when they have been positive. These results relates to our third proposition, which we discuss in the following section.

Table 4: Relationship to lagged flows

Columns 1 to 3 in Table 4 reports results for the following regression:

$$\gamma_t^F = \beta_0 + \beta_1 \times FFP_t^U + \beta_2 \times \sum_{s=t-3}^{t-1} FFP_s^U + \beta_3 \times \sum_{s=t-6}^{t-4} FFP_s^U + \dots + \beta_6 \times \sum_{s=t-18}^{t-16} FFP_s^U + \varepsilon_t$$

... where γ_t^F is the fund flow risk premium in month t and FFP_t^U is the unexpected fund flow to hybrid and equity mutual funds in month t . γ_t^F is denoted as the expected increase in a stock's excess return (in percentage points) for every one standard deviation increase in its FFB, holding the loadings on the Fama-French-Carhart factors constant. FFP_t^U has been normalized by its standard deviation. Columns 2 and 3 report results only for periods when the cumulative unexpected flow over the previous 18 months were negative and positive, respectively. Columns 4-6 report the average risk premium on the fund flow factor for the full sample, when unexpected flows the previous 18 months have been negative, and when unexpected flows the previous 18 months have been positive, respectively. Robust standard errors were used to adjust for heteroscedasticity. t-stats are reported in parentheses.

	Dependent variable: γ^F					
	Full Sample (1)	Neg. Past Flows (2)	Pos. Past Flows (3)	Full Sample (4)	Neg. Past Flows (5)	Pos. Past Flows (6)
Fund flows _{t} (unexpected)	0.905*** (4.66)	1.059*** (3.91)	0.516*** (5.08)			
Lagged ($t - 3$ to $t - 1$)	-0.058 (-0.27)	-0.026 (-0.07)	-0.649** (-2.41)			
Lagged ($t - 6$ to $t - 4$)	-0.542* (-1.79)	-0.961* (-1.79)	-0.100 (-0.46)			
Lagged ($t - 9$ to $t - 7$)	-0.643* (-1.86)	-1.197* (-1.88)	-0.159 (-0.81)			
Lagged ($t - 12$ to $t - 10$)	-0.420* (-1.78)	-1.091** (-2.28)	0.043 (0.23)			
Lagged ($t - 15$ to $t - 13$)	-0.340** (-1.97)	-0.977** (-2.29)	-0.169 (-0.70)			
Lagged ($t - 18$ to $t - 16$)	0.055 (0.36)	-0.448 (-1.45)	0.139 (0.58)			
Constant	0.256** (2.38)	-0.257 (-0.96)	0.115 (0.56)	0.233** (2.04)	0.472** (2.39)	-0.021 (-0.20)
Observations	315	162	153	315	162	153
R-squared	0.263	0.345	0.172	0.000	0.000	0.000

4.2.3 Test of proposition 3 (asymmetric price pressure)

Let us elaborate on the findings reported in Table 4. As we saw above, the risk premium on the fund flow factor is negatively correlated with previous flows, and this result is stronger when flows have been negative. In fact, if we look at columns 5 and 6, we see that the average fund flow risk premium is significantly positive when past flows have been negative, while it is negative but insignificant for periods where past flows have been positive. Furthermore, the reversal effect seems to be much more prominent when past flows have been negative, as indicated by the higher coefficient on contemporaneous unexpected flows in that scenario. The reversal is also slower when past flows have been negative, since we observe a negative correlation as far back as five quarters before the observation date, compared to only one quarter back when flows have been positive. Finally, the explanatory power of lagged and contemporaneous flows on the fund flow risk premium is higher when past flows have been negative. This all aligns with our third proposition: The price pressure is asymmetric.

Figure 4 sheds more light on the reversal patterns in response to different degrees of flow. The graphs plot the risk premium on the fund flow factor for each month relative to the observation date. Each observation is grouped into one of six groups depending on the unexpected flows in the quarter preceding the observation date. For example, the bottom 10% graph only includes observations where the unexpected flows in the preceding quarter are among the lowest 10% in the sample.

As we can see, the reversal pattern only exists for the bottom 10% group. Interestingly, the top 10% group shows a positive risk premium leading up to the observation date, but no reversal afterwards. The middle groups all show little or no reversal patterns, indicating that it is the extreme fund flows that are affecting the pricing of fund flow risk the most.

Compare the bottom 10% and the top 10% graphs in Figure 4 to the schematic overview in Figure 2.B. We then see that these patterns correspond relatively well to the lower and upper plots respectively—when flows have been negative we observe a negative risk premium followed by a reversal and when flows have been positive we observe the opposite, but the effect is lower in magnitude.

In sum, these findings strongly support our third proposition and we have thus found an explanation to the positive risk premium on the fund flow factor. In the next section, we illustrate how these findings can be used to construct a trading strategy based on fund flows.

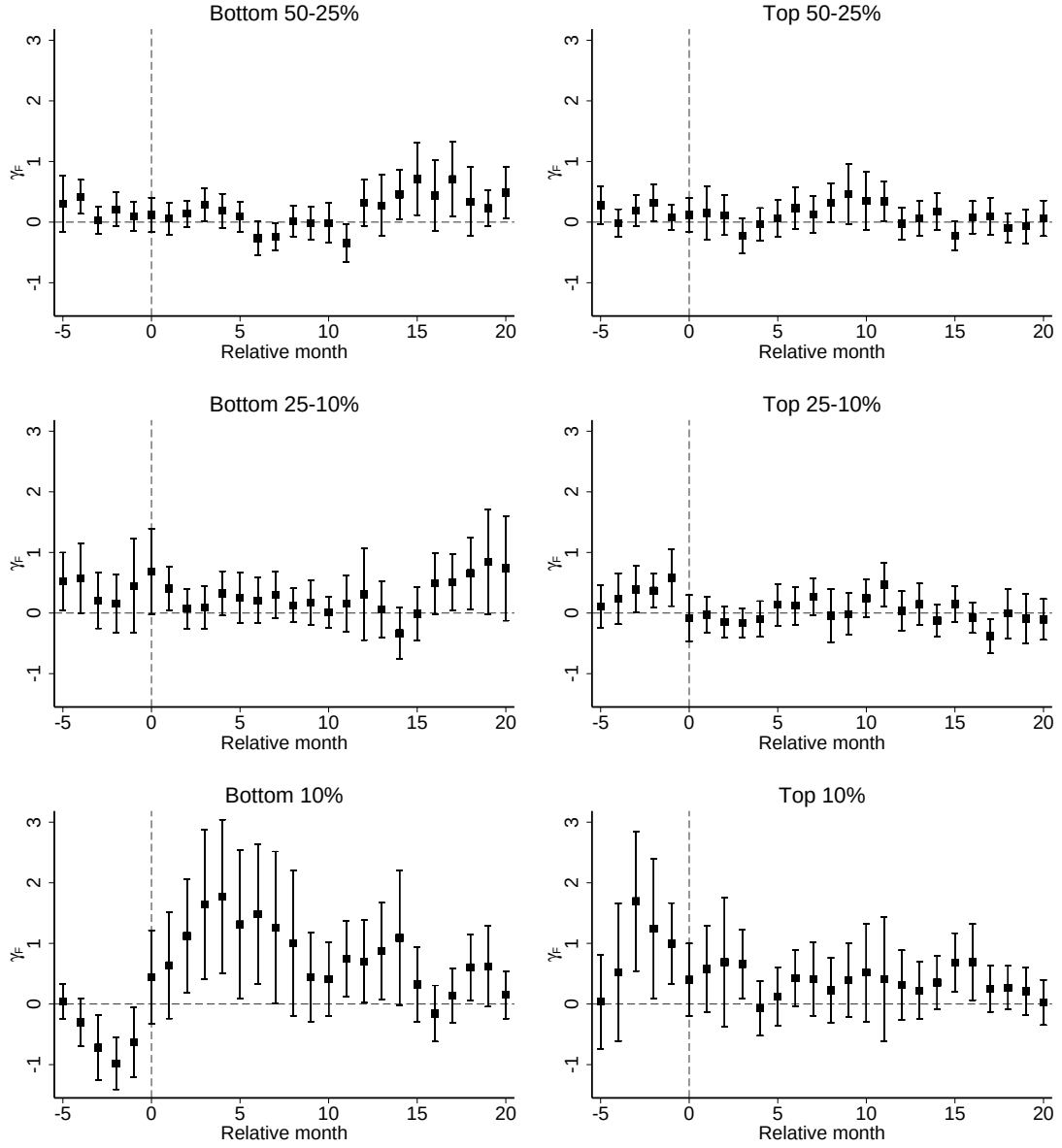


Figure 4: Reversal patterns in response to different degrees of flow

The graphs plot the risk premium on the fund flow factor and its 90% confidence interval for each month relative to the observation date. Each observation is grouped into one of six groups, based on the unexpected fund flow in the preceding quarter: bottom 10%, bottom 10–25%, bottom 25–50%, top 50–25%, top 25–10% and top 10%. The unexpected fund flow is the residual in an AR(3) model predicting fund flows based on lagged fund flows. γ_t^F is the expected increase in a stock's excess return, denoted in percent, for every one standard deviation increase in its FFB, holding the loadings on the Fama-French-Carhart factors constant.

4.2.4 Trading strategy

Section 3 showed that the premium on the fund flow factor is positively priced on average, and Section 4.2.2 showed that it is negatively related to past flows. What has yet to be demonstrated is the economic significance of these findings. To this end, this section tests a simple, long-short strategy that takes directional bets on the fund flow factor based on the pattern of past flows. The specification and performance of the strategy are detailed below.

Specification

Each month, we sort stocks into equal-weighted decile portfolios based on their fund flow beta. These portfolios are referred to below as portfolio 1 (comprising of the lowest-FFB stocks) through 10 (comprising of the highest-FFB stocks). The strategy has two phases:

Phase 1, “Active”: The strategy is long portfolio 10 and short portfolio 1.

Phase 2, “Passive”: The strategy is passive; it is not long or short in any portfolio (thus yielding a return of zero).

The phase at month t is determined by the following ruleset:

$$\text{phase}_t = \begin{cases} \text{“Active”} & \text{if } \sum_{s=t-18}^{t-1} FFP_s^U < 0 \\ \text{“Passive”} & \text{otherwise} \end{cases}$$

This ruleset is motivated by the results from Section 4.2.3 and, specifically, columns 5–6 of Table 4, which demonstrated that the premium on the fund flow factor is negatively associated with the direction of past flows.¹²

Performance

Consider the following (generalized) multi-factor model:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i^0 \times FCT_t^0 + \dots + \beta_i^j \times FCT_t^j + \varepsilon_{i,t}$$

... where $R_{i,t}$ is the return of stock i in month t , $R_{f,t}$ is the risk-free rate in month t , and $FCT_t^0, \dots, FCT_t^j$ are the return on $j + 1$ factor portfolios in month t .

¹²The rules have deliberately been made very simple in order to increase the robustness of our results. Alternatively, we could have based the rules on more precise prediction models, similar to those in columns 2–3 of Table 4. However, it is impossible to do so without either introducing look-ahead bias or severely limiting the sample period of the performance tests.

Let R_l be the return on the portfolio that the strategy is long and R_s be the return on the portfolio that the strategy is short. Then, the population model of the excess return on the strategy is:

$$\begin{aligned} R_{l,t} - R_{s,t} &= (\alpha_l - \alpha_s) + (\beta_l^0 - \beta_s^0) \times FCT_t^0 + \dots \\ &\quad + (\beta_l^j - \beta_s^j) \times FCT_t^j + (\varepsilon_{l,t} - \varepsilon_{s,t}) \\ &= \alpha^* + \beta^{*0} \times FCT_t^0 + \dots + \beta^{*j} \times FCT_t^j + \varepsilon_t^* \end{aligned} \quad (14)$$

Table 5 reports the estimated parameter values of (14) for three different return models: the CAPM, the Fama-French three-factor model, and the Fama-French-Carhart four-factor model. As can be seen from columns 1–3, the $\hat{\alpha}^*$ are positive and significantly different from zero on *at least* the 10% significance level across all three specifications when we run the regressions on both phases simultaneously. Thus, we can tentatively reject the null hypothesis that the strategy does not generate abnormal returns, i.e., that $\alpha_l = \alpha_s \Leftrightarrow \alpha^* = 0$.

The coefficients in columns 1–3 should be interpreted with some care, however, as the betas are phase-dependent. In other words, the expected return on the strategy need not be linear in factor returns, which could make the $\hat{\alpha}^*$ in columns 1–3 somewhat difficult to interpret. Therefore, to make sure that we do not make any false inferences, columns 4–6 reruns the regressions on only the Active phase of the strategy. The $\hat{\alpha}^*$ in these columns are of similar magnitude¹³ and significance as the $\hat{\alpha}^*$ in columns 1–3, thus confirming our previous results.

As an additional performance appraisal, Figure 5 plots the excess return and realized alpha of the strategy over time. Referring back to Figure 1, we note two things: First, not surprisingly, the payoff of the strategy closely tracked the premium on the fund flow factor over the sample period. Second, the strategy was in its passive phase for most of the 1990s, indicating that fund flows followed an upward trend during these years. Consequently, flow-induced price pressures were presumably only very marginal throughout this period, which might explain why the premium on the fund flow factor averaged close to zero at the time.

In sum, we conclude that it is possible to construct a very simple long-short strategy that capitalizes on the premium on the fund flow factor and thus yields abnormal returns. Further, the abnormal returns appear to be of quite large economic significance, with predicted monthly alphas ranging from 0.33% to 0.54% depending on which return model that is used.

¹³The average four-factor $\hat{\alpha}^*$ is now $0.911\% \times \frac{162}{332} \approx 0.48\%$. It is thus marginally lower than the $\hat{\alpha}^*$ in column 3, suggesting that the return on the strategy is slightly convex with respect to factor returns.

Table 5: Performance and risk exposures of the trading strategy

This table shows the performance and risk exposures of the strategy using three different return models: the CAPM, the Fama-French three-factor model, and the Fama-French-Carhart four-factor model. The strategy is as follows: Stocks are sorted into equal-weighted decile portfolios based on their fund flow beta each month. The strategy has two phases: (1) “Active” and (2) “Passive”. In the Active phase, the portfolio is long the 10th decile portfolio and short the 1st decile portfolio. In the passive phase, the strategy is not long or short in any portfolio (and thus yields a return of zero). The phase at month t is determined by the following ruleset: The strategy is in the Active phase if the cumulative unexpected fund flow over the prior 18 months has been negative. Otherwise, it is in the Passive phase. The unexpected fund flow is the residual in an AR(3) model predicting fund flows based on lagged fund flows. Coefficients are denoted in percent. Robust standard errors were used to adjust for heteroscedasticity. t-stats are reported in parentheses.

	Dependent variable: $R_l - R_s$					
	Phase: Active or Passive			Phase: Active		
	(1)	(2)	(3)	(4)	(5)	(6)
M	0.101*	0.097	0.024	0.181*	0.176*	-0.022
	(1.80)	(1.42)	(0.38)	(1.82)	(1.69)	(-0.20)
SMB		0.011	0.033		0.033	-0.063
		(0.19)	(0.59)		(0.21)	(-0.49)
HML		-0.014	-0.101		-0.104	-0.313
		(-0.12)	(-0.86)		(-0.40)	(-1.41)
UMD			-0.261***			-0.457***
			(-4.03)			(-5.20)
Constant	0.327*	0.332*	0.538**	0.662*	0.631*	0.911***
	(1.87)	(1.69)	(2.50)	(1.85)	(1.91)	(2.76)
Observations	332	332	332	162	162	162
R-squared	0.022	0.022	0.173	0.039	0.043	0.284

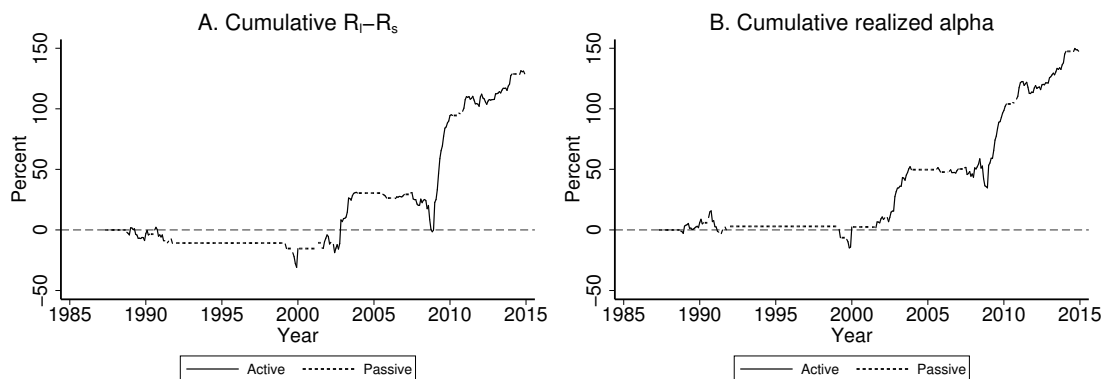


Figure 5: Performance of the trading strategy over time

This figure plots the performance of the trading strategy over time. The strategy is as follows: Stocks are sorted into equal-weighted decile portfolios based on their fund flow beta each month. The strategy has two phases: (1) “Active” and (2) “Passive”. In the Active phase, the portfolio is long the 10th decile portfolio and short the 1st decile portfolio. In the passive phase, the strategy is not long or short in any portfolio (and thus yields a return of zero). The phase at month t is determined by the following ruleset: The strategy is in the Active phase if the cumulative unexpected fund flow over the prior 18 months has been negative. Otherwise, it is in the Passive phase. The unexpected fund flow is the residual in an AR(3) model predicting fund flows based on lagged fund flows. Panel A plots the cumulative excess return of the strategy. Panel B plots the cumulative realized alpha of the strategy. The realized alpha is defined as the difference between the strategy’s realized return and its expected return based on the Fama-French-Carhart four-factor model using phase-dependent betas. The y-axes are denoted in percent.

Relationship to UMD

While outside the scope of this paper, one interesting observation with regard to the risk characteristics of the strategy deserves mention: From column 6 in Table 5, high-FFB stocks appear to load very negatively on the UMD factor. This observation suggests that the fund flow factor and the momentum factor may be related. This is, to some extent, expected: In the Active phase, high-FFB stocks have been exposed to negative price pressures, therefore lowering their prior 12-month return. As a result, they are more likely to be sorted into—and thus correlate with—the “Down” portfolio (which the UMD factor is short) than the “Up” portfolio (which the UMD factor is long), hence giving them a negative loading on the UMD factor. Interestingly, this implies that the UMD factor should perform relatively poorly following downturns in flow, as the Down portfolio then may include underpriced, high-FFB stocks, whose prices are at risk of reverting. This also appears to be the case: In results not shown, the return on the UMD factor and its three-factor alpha are significantly lower (and insignificantly different from zero) when the cumulative unexpected fund flow over the prior 18 months is negative.

Taking the analysis one step further, these observations may offer a candidate explanation for the pricing of the UMD factor: The UMD factor may—at least in part—capture the momentum in price reversals of stocks that have previously been exposed to negative, flow-induced price pressures. Thus, whereas such stocks will load highly on the fund flow factor immediately following said price pressure, they will load highly on the UMD factor when flows have been non-negative in the previous 12 months and their prices hence have started to revert.¹⁴ As a result, high-FFB stocks will perform well following negative past flows whereas high-UMD stocks will perform well following positive past flows, which is exactly what we observe in the data.

5 Conclusions

This study set out to investigate the pricing of fund flow risk in order to develop the existing literature regarding the effect of institutional trading on asset prices. Our main result is that stocks that comove with mutual fund flows earn higher abnormal returns. In trying to explain this finding, we find that the rational hypothesis—that mutual fund managers underweigh stocks with exposure to fund flow risk—is not supported by the data. Instead, this phenomenon can be explained by a hypothesis based on assumptions of an inefficient market. Specifically, the positive risk premium on the fund flow factor comes from a reversal effect that arises due to flow-induced price pressure from mutual funds. Finally, this reversal effect is largest when there has been large unexpected shocks to the economy.

Some implications from our findings follow. Most important, our study, like the works from e.g. Brennan (1993) and Cornell and Roll (2005), shows the importance of taking institutional trading into account when developing asset pricing models. Second, unlike their works, it shows that a fully rational view is not sufficient when modeling the effect of institutional investors. Instead, care must be taken to implement the fact that institutional traders are subject to constraints, such as the risk of sudden redemptions, that may force them to act irrationally. As we have seen, an investor that exploits this irrationality will earn an average monthly alpha of 0.5%, which further emphasizes the economic importance of our findings.

On a grander scale, this paper is about the pricing of different sources of risk in the economy. Critically, our results suggest that fund flow risk is *not* a priced risk factor in the sense that fund flows are partially correlated with the

¹⁴Since reversals take longer than 12 months, as suggested in Sections 4.2.2–4.2.3, the premium on the UMD factor will still be positive.

marginal utility of any market participant. Rather, it is an expression of market inefficiencies stemming from flow-induced price pressures. Consequently, it differs from traditional risk factors, like market risk, in the critical sense that an asset that loads highly on the fund flow factor should not be discounted at a higher rate than an otherwise identical asset that has a lower loading on the fund flow factor. Instead, rational investors should value both assets at the same dollar value. Hence, while it would be possible to augment current multi-factor models with a factor portfolio that proxies for “fund flow risk” (e.g., the return on the fund flow strategy described in Section 4.2.4), it would be inappropriate to do so if the purpose of the model is to estimate the required rate of return on an investment opportunity.¹⁵

The importance of understanding which risk factors that we should care about cannot be overstated. Surely, one would allocate one’s wealth very differently if cash streams were discounted using required rates of return that were derived from the CAPM compared to if they were derived from a multi-factor model. Therefore, the welfare consequences of mistaking a *pattern in mispricings* (e.g., that high-FFB stocks on average are underpriced because of flow-induced price pressures) for a *priced risk factor* (e.g., that high-FFB stocks have a lower price than other stocks because open-end funds rationally underweigh stocks that comove with fund flows) are tremendous—not only for individual investors but also for the economy as a whole.

Thus, we believe that future research should aim to continue to improve our understanding of, and untangle the precise relationships between, the many proposed but yet to be explained risk factors in the economy. For example, we saw in Section 4.2.4 some evidence that the momentum factor could be related to fund flow risk.¹⁶ Further, it is also conceivable that liquidity risk could be related to fund flow risk, as fund flows should correlate well with changes in market-wide liquidity. Curiously, Pástor and Stambaugh (2003) suggest that the liquidity and momentum risk premia may be related. In extension, it could be that fund flow risk, momentum risk, and liquidity risk all proxy for the same basic phenomenon. If this is the case, and if the price pressure hypothesis holds, it could imply that neither liquidity nor momentum risk—similar to fund flow risk—should be priced risk factors to rational investors.

¹⁵On the other hand, if the purpose is to evaluate the *performance* of an investment portfolio, the argument becomes less clear-cut; given that it would be trivial to implement a fund flow strategy—similar to the one described in Section 4.2.4—as a part of the portfolio, it may seem appropriate to also control for the additional performance that such a strategy would yield.

¹⁶We are not the first to suggest this connection; a relationship between flow-induced price pressure and momentum has previously been proposed by, for example, Lou (2012).

To conclude, we hope that this paper has not only introduced yet another risk factor but also offered a viable explanation for why it appears to be priced—and why rational investors should not let it affect their investment decisions.

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